The Application of Logistic Regression to Examine the Predictors of Car Crashes Caused by Alcohol

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CPLN 671/MUSA 500: Statistical and Data Mining Methods for Urban Data Analysis

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20 April 2025

Introduction

This report examines a logistic regression model on alcohol-induced car crashes across Philadelphia, PA. According to the United States Department of Transportation, someone dies in a motor vehicle crash involving an alcohol-impaired driver every 51 minutes. The direct impact on the lives of those who know the victims and the externalities (valued at $59 billion in economic impact) that society bears from the cascading effects of alcohol-related crashes make this a critical study.

In this study, the following predictors used in this regression:

- Crash-related Predictors:
  - Whether the accident resulted in a fatality (FATAL_OR_M)
  - If a vehicle was overturned in the crash (OVERTURNED)
  - If a cell phone was used while driving (CELL_PHONE)
  - If a car was speeding (SPEEDING)
  - If the crash included aggressive driving (AGGRESSIVE)
- Age-Related Predictors:
  - If the driver was 16 or 17 years of age (DRIVER1617)
  - If the driver was 65 years old or older (DRIVER65PLUS)
- Census-related Predictors:
  - Percentage with at least bachelor's degree (PCTBACHMOR)
  - Median household income (MEDHHINC)

The predictors above that are used in the logistics regression model likely capture the behavioral mechanisms and socioeconomic factors associated with alcohol-induced crashes. Crash-related variables such as FATAL_OR_M, OVERTURNED, SPEEDING, and AGGRESSIVE reflect how alcohol changes judgment and motor control, increasing a crash's severity and reckless behaviors. CELL_PHONE may indicate compounded risk-taking through prohibited cell phone use while driving. Age-related predictions (DRIVER1617 & DRIVER65PLUS) account for vulnerable populations who can legally drive, including inexperienced teen drivers and older adults with slower reaction times and alcohol metabolism. The census variables (PCTBACHMOR & MEDHHINC) may indirectly influence crash rates; higher incomes and educational attainment may correlate with greater DUI awareness and better financial access to alternative transportation options. Lower-income areas may have limited transportation options and a higher density of alcohol outlets. In unison, each of these predictors can help depict alcohol's effects from a socioeconomic and behavioral context that shape crash risk.

The R Programming Language was used to run the entire logistic regression for this analysis using RStudio.

Methods

Binary Dependent Variables and OLS Regressions

Logistic regression models are used since the outcome variable of car alcohol-induced is a binary variable, meaning the value can only be 0 or 1.

An ordinary least squares (OLS) regression assumes continuous, normally dependent variables ranging from negative to positive infinity. However, when the outcome is binary, an OLS faces several issues. Since an OLS assumes normally distributed errors with constant variance, binary outcomes in the model produce residuals that are non-normal and heteroscedastic. Additionally, OLS works to predict probabilities outside of a [0,1] range, which is not possible for a binary outcome. Finally, there is inherently a nonlinear relationship between predictors and a binary outcome – doubling the income doesn't double the chance of a crash.

How Logistic Regressions Work

The concept of odds is discussed first to introduce how logistic regressions work. Odds are the ratio of the probability of an event occurring to it not occurring.

$$Odds = \frac{\#\, desirable\ outcomes}{\#\, undesriable\ outcomes} = \frac{P(Y=1)}{P(Y\neq 1)}$$

$$= \frac{P(Y=1)}{P(Y=0)} = \frac{P(Y=1)}{1-P(Y=1)} = \frac{p}{1-p}$$

where

$$p = \frac{\#\, desirable\ outcomes}{\#\, possible\ outcomes} = P(Y=1)$$

The odds ratio compares the odds of an event occurring in one group to the odds that the same event occurs in another group. A ratio greater than 1 indicates higher odds in the first group, whereas an odds ratio less than 1 indicates lower odds for the first group.

Since logistic regression deals with a binary outcome, the logistic regression models log-odds compared to probabilities. Log-odds are derived from the natural logarithm of the odds:

$\ln\left(\frac{p}{1-p}\right)$. This value is also called the logit. In logit form, the logistic regression expresses the log-odds event as a linear combination of predictors, written as:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \mathrm{FATAL}_{\mathrm{OR_M}} + \beta_2 \mathrm{OVERTURNED} + \beta_3 \mathrm{CELL}_{\mathrm{PHONE}} + \beta_4 \mathrm{SPEEDING} + \beta_5 \mathrm{AGGRESSIVE} + \beta_6 \mathrm{DRIVER1617} + \beta_7 \mathrm{DRIVER65PLUS} + \beta_8 \mathrm{PCTBACHMOR} + \beta_9 \mathrm{MEDHHINC}$$

where

- p: probability that the car crash is alcohol-induced
- β_0: intercept term representing the log-odds of the accident being alcohol-induced occurring when all predictors are zero
- $\beta_1, \beta_2, \beta_3 \ldots \beta_9$: coefficient representing the change in the log-odds of the accident involving a drunk driver for a one-unit increase in each predictor, holding others constant
- $\mathrm{FATAL}_{\mathrm{OR_M}}$: if the accident was fatal
- $\mathrm{OVERTURNED}$: if the car was overturned
- $\mathrm{CELLPHONE}$: if the driver was using their cellphone
- $\mathrm{SPEEDING}$: if the car was speeding
- $\mathrm{AGGRESSIVE}$: if aggressive driving was involved
- $\mathrm{DRIVER1617}$: if the driver was 16 or 17 years old
- $\mathrm{DRIVER65PLUS}$: if the driver was at least 65 years old

- PCTBACHMOR: percent of residents with at least a bachelor's degree in the census tract where the car crash took place
- MEDHHINC: median household income in the census tract where the car crash took place

The logit $\ln\left(\frac{p}{1-p}\right)$ represents the natural logarithm of the odds of the event occurring, taking any value from negative infinity to positive infinity. Since probabilities only fall between 0 and 1, the logistic regression includes a logistic function that transforms log-odds into probabilities. The logistic function (the inverse logit) is expressed for a singular predictor variable below.

$$p = \frac{e^{\beta_0+\beta_1 x_1}}{1+e^{\beta_0+\beta_1 x_1}} = \frac{1}{1+e^{-\beta_0-\beta_1 x_1}}$$

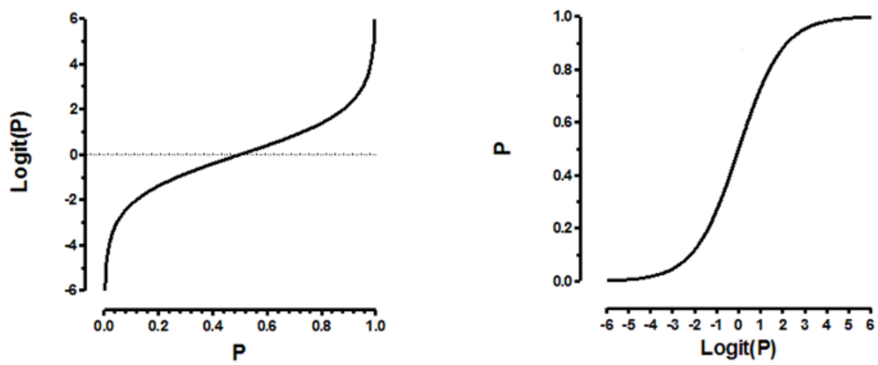
Therefore, the exhaustive equations for this analysis with multiple predictors for this study are below:

$$p = \frac{e^{\beta_0+\beta_1 \mathrm{FATAL_{OR_M}}+\beta_2 \mathrm{OVERTURNED}+\beta_3 \mathrm{CELL_{PHONE}}+\beta_4 \mathrm{SPEEDING}+\beta_5 \mathrm{AGGRESSIVE}+\beta_6 \mathrm{DRIVER1617}+\beta_7 \mathrm{DRIVER65PLUS}+\beta_8 \mathrm{PCTBACHMOR}+\beta_9 \mathrm{MEDHHINC}}}{1+e^{\beta_0+\beta_1 \mathrm{FATAL_{OR_M}}+\beta_2 \mathrm{OVERTURNED}+\beta_3 \mathrm{CELL_{PHONE}}+\beta_4 \mathrm{SPEEDING}+\beta_5 \mathrm{AGGRESSIVE}+\beta_6 \mathrm{DRIVER1617}+\beta_7 \mathrm{DRIVER65PLUS}+\beta_8 \mathrm{PCTBACHMOR}+\beta_9 \mathrm{MEDHHINC}}}$$

$$= \frac{1}{1+e^{-(\beta_0+\beta_1 \mathrm{FATAL_{OR_M}}+\beta_2 \mathrm{OVERTURNED}+\beta_3 \mathrm{CELL_{PHONE}}+\beta_4 \mathrm{SPEEDING}+\beta_5 \mathrm{AGGRESSIVE}+\beta_6 \mathrm{DRIVER1617}+\beta_7 \mathrm{DRIVER65PLUS}+\beta_8 \mathrm{PCTBACHMOR}+\beta_9 \mathrm{MEDHHINC})}}$$

Figure 1

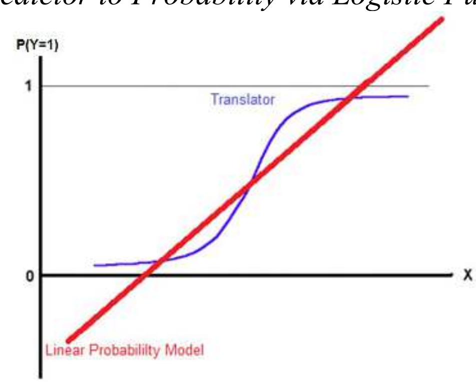
Plot of Logit Function with a Scaled Probit



The logistic function transforms the unbounded output of linear regression into valid probabilities between 0 and 1 using its characteristic S-shaped curve. As the predicted value from the linear regression model approaches -∞, the transformed probability approaches 0% (0.0). As the predicted value approaches +∞, the transformed probability approaches 100% (1.0). At a predictor value of 0, the probability equals exactly 50%.

Figure 2

Transformation from Linear Predictor to Probability via Logistic Function



The logistic regression coefficients can be exponentiated to produce odds ratios, which provide interpretable estimates of how each predictor variable affects the likelihood of a crash involving alcohol-impaired driving.

Hypotheses for Each Predictor

For each predictor x_i, the hypotheses are as follows:

- $Null\ Hypothesis\ (H_0): \beta_i = 0\ (OR_i = 1)$
  - This means the predictor has no effects on the log-odds of the dependent variable
- Alternative Hypothesis $(H_a): \beta_i \neq 0\ (OR_i \neq 1)$
  - This means the predictor has an effect on the log-odds of the dependent variable

The Wald Statistic assesses the statistical significance of individual logistic regression coefficients by testing the null hypothesis that the estimated parameter $\hat{\beta}_i$ equals zero. The Wald statistic for each coefficient is calculated as:

$$\frac{\hat{\beta}_i - E(\hat{\beta}_i)}{\sigma_{\hat{\beta}_i}} = \frac{\hat{\beta}_i - 0}{\sigma_{\hat{\beta}_i}} = \frac{\hat{\beta}_i}{\sigma_{\hat{\beta}_i}}$$

where

- $\hat{\beta}_i$ = estimated coefficient for the predictor variable
- $\sigma_{\hat{\beta}_i}$ = standard error of the estimated coefficient

Researchers use the Wald statistic to check for statistical significance when testing regression coefficients. This test assumes the coefficient equals zero under the null hypothesis (no effect). The Wald statistic follows a normal distribution, so analysts compare its value to standard normal critical values to compute a p-value. If this p-value falls below a pre-set threshold (like 0.05), the null hypothesis is rejected, suggesting the predictor has a meaningful effect.

While regression outputs provide raw coefficients (β), many researchers find odds ratios more intuitive. These are derived by exponentiating the coefficients, making them easier to interpret. For example, an odds ratio of 2.0 means the outcome becomes twice as likely for each unit increase in the predictor.

The Assessment of Quality of Model Fit

Several methods assess the quality of the logistic regression model fit.

R-Squared

R-squared is a standard metric for evaluating model quality in linear regression. However, due to the dependent variable, the interpretation of this statistic is less intuitive in logistic regressions. Several alternative measures exist, such as McFadden's R-squared, which compares the model's likelihood to a null model. Unlike in OLS, these values do not represent a percentage of variance accounted for by the model—their best use is to compare the relative fit of other models rather than assess absolute fit.

Akaike Information Criterion (AIC)

AIC is an estimator of the quality of a statistical model and the prediction error, measuring the trade-off between the model's complexity and goodness of fit. A lower AIC indicates a better model. The following equation represents AIC:

$$AIC = 2k - 2\ln \hat{L}$$

where

- k = the number of estimated parameters in the model
- $\hat{L}$ = the maximized value of the likelihood function for the model

k is a parameter that penalizes complexity, where overfit models with exhaustive parameters yield a larger AIC.

Error Metrics

In logistic regression, the fitted (predicted) values $\hat{y}$ represented an estimated probability that the dependent variable Y=1 given the predictor variables. This prediction comes from the logistic function of the linear combination of the predictors shown below.

$$\hat{y} = p = \frac{e^{\beta_0+\beta_1 x_1+\cdots+\beta_k x_k}}{1+e^{\beta_0+\beta_1 x_1+\cdots+\beta_k x_k}}$$

A probability threshold is chosen to classify an observation, such that if $\hat{y}$ is greater than a cut-off value, the model will predict Y-1, and if $\hat{y}$ is less than or equal to the cut-off value, the model predicts Y=0.

Table 1

Actual & Predicted Y

ACTUAL/PREDICTED Y	Predicted Negative (0)	Predicted Positive (1)
ACTUAL NEGATIVE (0)	*a (True Negative)*	*c (False Positive)*
ACTUAL POSITIVE (1)	*b (False Negative)*	*d (True Positive)*

The following section will recall the letters a, b, c, and d to introduce the error metrics.

Specificity measures the proportion of actual negatives correctly predicted by the model to be negative, which is calculated as $\frac{a}{a+c}$, which is known as a "True Negative Rate". A larger value indicates better identification of positive cases.

Sensitivity is the proportion of actual positives correctly predicted by the model to be true, which is calculated by $\frac{d}{b+d}$. This is known as a "True Positive Rate". A higher sensitivity indicates better identification of positive cases.

A misclassification rate represents the proportion of incorrect predictions made by a model, which is $\frac{b+c}{a+b+c+d}$. A lower rate indicates better overall predictive performance.

In logistic regression, the cut-off probability used to classify observations as positive (Y=1) or negative (Y=0) influences key evaluation metrics like sensitivity, specificity, and misclassification rate. While a default threshold of 0.5 is often applied—where probabilities above 0.5 are classified as Y=1 and those below as Y=0—adjusting this value can help balance model performance based on specific needs.

For instance, raising the threshold (e.g., to 0.7) increases specificity but reduces sensitivity, as the model becomes more conservative in predicting positives. Conversely, lowering the threshold (e.g., to 0.3) boosts sensitivity but reduces specificity, making the model more permissive in assigning the positive class. However, an excessively high or low threshold may lead to a surge in false negatives or false positives, worsening the misclassification rate. By evaluating different thresholds, practitioners can fine-tune the model's performance to align with its intended application.

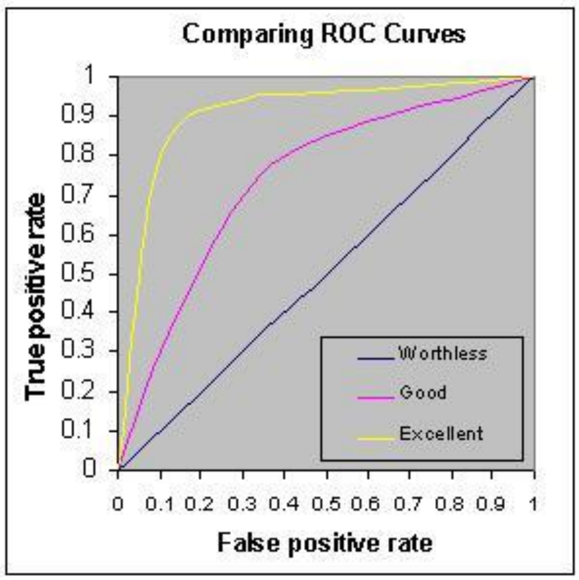
Receiver Operating Characteristic (ROC) Curve

An ROC curve is a graphical representation of a model's classification performance composed of the True Positive Rate (sensitivity) plotted on the y-axis and the False Positive Rate

(1 - Specificity), on the x-axis, at different probability cut-offs. This illustrates trade-offs between performance metrics across different thresholds.

A perfect classification would contain a point in the upper-left corner, demonstrating 100% sensitivity and 0% false positives. However, a perfect fit could indicate an overfit model that may not work well on a new set of data compared to the existing set of data. A well-balanced ROC curve indicates a strong performance without overfitting.

Figure 3
Example of an ROC Curve



Several methods identify optimal probability cut-offs based on ROC curves. Youden's Index is where the optimal cut-off is the threshold that maximizes (Sensitivity + Specificity - 1).

This analysis determined the optimal classification threshold by minimizing the Euclidean distance from the ideal point (0,1) on the ROC curve, where sensitivity reaches 1 (perfect true positive rate) and the false positive rate is 0. This approach identifies the point where both sensitivity and specificity are jointly maximized.

The Area Under the ROC Curve (AUC) is a key measure of the model's overall discriminative power, aiding in feature selection and model comparison. It reflects the model's ability to differentiate between positive and negative cases, with values ranging from 0.5 to 1.0:

- An AUC of 0.5 implies no predictive ability (equivalent to random guessing).
- An AUC near 1.0 suggests near-perfect classification.

Thus, a stronger model typically achieves an AUC significantly above 0.5, with higher values indicating better performance. Value ranges for the area under the curve (AUC) are as follows.

- Excellent: .90-1
- Good: .80-.90
- Fair: .70-.80
- Poor: .60-.70
- Fail: .50-.60

Each is a conservative estimate, and some statisticians argue that an AUC above 0.7 is sufficient.

Assumptions of Logistic Regression

Assumptions for OLS	Assumptions for Logistic Regression
• **Independence of observations**	• Independence of observations
• **A linear relationship between the dependent and each independent variable**	• A linear relationship between the log-odds of the dependent variable and each independent variable
• **Normality of residuals**	• No assumption of normality of residuals
• **Homoscedasticity**	• No assumption of homoscedasticity
• **No multicollinearity**	• No severe multicollinearity
• **At least about 10 per predictor are**	• At least 50 observations per predictor are

needed	needed
	• A dependent variable must be binary

An assumption that the independence of observations in OLS is also true for logistic regression. Although a linear relationship between the dependent and independent variables is not required, linear relationships between the log-odds of Y = 1 are assumed to be linearly related to each independent variable. Logistic regression does not assume homoscedasticity since the variance of the errors is not constant; the errors do not need to be normal either. For linear regressions, severe multicollinearity should not be present for independent variables. A large sample of at least 50 observations per predictor is required because a maximum likelihood estimation is conducted, rather than least squares, which is used to estimate regression coefficients.

Exploratory Analyses Before Logistic Regressions

Before conducting logistic regression, cross tabulations and chi-square tests are relevant analyses to identify relationships between the dependent variable and predictor values.

Cross-tabulations summarize the relationship between the binary dependent variable and the binary predictors. In this study, the binary dependent variable is whether a car crash was alcohol induced, whereas the predictors include speeding and aggressive driving, among others.

A chi-square test examines if there is a statistically significant association between two categorical variables. The test's null hypothesis (H_0) is that there is no association between the two variables, and they are independent. The alternative hypothesis (H_a) is an association between the two variables, which depend on one another. Suppose the p-value from the chi-square test is below the significance level. In that case, the null hypothesis is rejected, indicating a potential relationship between the dependent and independent variables, making conducting a logistic regression meaningful.

The average values across the two groups are compared for continuous variables, such as percentage with at least a bachelor's degree or median household income. An independent samples t-test confirms if the average values of continuous predictors significantly differ between crashes that were alcohol-induced or otherwise. This test's null hypothesis (H_0) is that the means of the two groups are equal, whereas the alternative hypothesis (H_a) is that the means are not equal.

Suppose the p-value from the t-test is below the significance level. In that case, the null hypothesis is rejected, demonstrating a significant difference in the means of each independent variable across the two categories of the dependent variable.

Cross-tabulations and t-tests help identify useful predictors associated with the outcome variable to confirm meaningful variable selection for a logistic regression.

Results

Exploratory Analysis Results

An initial analysis of Philadelphia's crash dataset (N = 43,364) revealed that 5.73% (2,485 crashes) of all crashes involved alcohol impairment, whereas all remaining crashes did not (94.27%, or 40,879 total crashes). One in every 17 crashes in Philadelphia involves alcohol as a contributing factor, representing a share of preventable crashes.

Table 2 below demonstrates a cross-tabulation for each binary predictor with alcohol involvement, including chi-square test statistics.

Table 2

Cross-Tabulation of all Predictors of Alcohol Involvement

	No Alcohol Involved (DRINKING_D=0)		**Alcohol Involved (DRINKING_D=1)**		**Total**	**χ2 p-value**
	N	**%**	**N**	**%**	**N**	**N**
FATAL_OR_M: Crash resulted in fatality or major injury	1181	2.889%	118	7.565%	1369	2.522202e-38
OVERTURNED: Crash involved an overturned vehicle	612	1.497%	110	4.427%	722	4.754231e-12
CELL_PHONE: Driver was using cell phone	426	1.042%	28	1.127%	454	0.6528070
SPEEDING: Crash involved speeding car	1261	3.085%	260	10.463%	1521	6.264010e-65
AGGRESSIVE: Crash involved aggressive driving	18522	45.309%	916	36.861%	19438	0.0214204
DRIVER1617: Crash involved at least one driver who was 16 or 17 years old	674	1.649%	12	0.483%	686	0.5586704
DRIVER65PLUS: Crash involved at least one driver who was at least 65 years old	4237	10.365%	119	4.789%	4356	0.009172028

The mean values of continuous predictors, PCTBACHMOR (percentage with at least a bachelor's degree) and MEDHHINC (median household income), were tabulated across crashes involving alcohol and those that didn't, with a summary depicted below in Table 2.

Table 3
Means of Two Continuous Predictors

	No Alcohol Involved		Alcohol Involved		t-test p-value
	Mean	**SD**	**Mean**	**SD**	**N**
PCTBACHMOR: % with bachelor's degree or more	16.56986	18.21426	16.61173	18.72091	0.9137
MEDHHINC: Median household income	31483.05	16930.1	31998.75	17810.5	0.16

Key Insights

Completing an analysis showed clear associations between alcohol involvement and specific crash characteristics, in addition to driver demographics, showing both positive and negative relationships.

Although only 2.9% of crashes not involving alcohol resulted in either fatal or major injuries, the percentage of crashes with this severity of casualties was 2.6 times higher if the driver was alcohol impaired, at 7.6%. This difference indicates a connection between crash severity and alcohol impairment. Additionally, crashes with overturned vehicles indicated a significant positive association with alcohol impairment. Speeding is the strongest predictor in an alcohol-impaired accident, with 10.5% of crashes involving alcohol compared to only 3.1%.

In contrast, crashes involving aggressive driving and age-related predictors demonstrated negative relationships with alcohol involvement, as alcohol impairment was less likely for each predictor. Both age predictors (16-17 & 65+) also demonstrated negative associations, as alcohol impairment was significantly lower among these age groups, indicating these demographics are less likely to be part of alcohol-related crashes.

Cell phone use and alcohol impairment showed no significant relationship, suggesting that distracted driving caused by cell phone use may be independent of alcohol impairment as a contributing factor to all crashes in Philadelphia.

Both census variables (percent with a bachelor's degree or higher and median household income) had no statistically significant difference in their mean values across all crashes (regardless of alcohol) based on the t-test, as the p-value was over 0.05 for both variables.

Overall, the strongest connections to alcohol-impaired crashes are observed with speeding and crash severity, such as fatalities or major injuries. In contrast, aggressive driving and age-related factors are associated with little alcohol involvement.

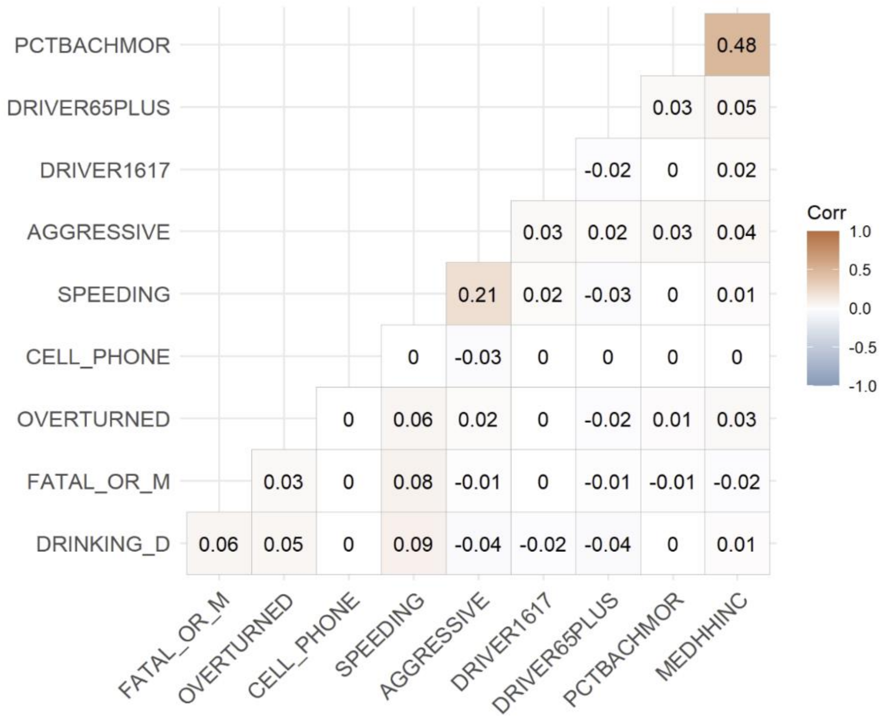
Analysis of Logistic Regression Assumptions

Each remaining assumption of logistic regression beyond discovering correlations requires descriptive diagnostics tests and theoretical evaluations.

Multicollinearity

Assessing multicollinearity required a Pearson correlation matrix to generate diagnostics for all binary and continuous predictors.

Figure 4
Correlation Matrix of Variables



A Pearson correlation measures the linear relationship between two continuous variables. However, this can be misleading for binary variables. Pearson correlation assumes a linear relationship, which binary variables may not follow. Additionally, the correlation values between binary variables represent association, but it is less intuitive compared to chi-square tests. Higher correlations between binary predictors may still underestimate relationships in irregular contexts, such as a rare event.

In this study, the correlations across all predictors remained low, with the highest correlation between the two continuous variables, PCTBACHMOR and MEDHHINC, at 0.48. These two factors are only moderately correlated, indicating there is no severe collinearity between predictors. To add depth to this analysis, the variance inflation factor (VIF) was calculated for each predictor, shown below in Figure 5. All VIF values are below 2, confirming no concern for multicollinearity.

Figure 5

VIF Values from RStudio

```
  FATAL_OR_M   OVERTURNED   CELL_PHONE     SPEEDING   AGGRESSIVE   DRIVER1617
    1.015556     1.010465     1.000624     1.234130     1.214620     1.000847
DRIVER65PLUS   PCTBACHMOR     MEDHHINC
    1.003152     1.302936     1.308495
```

Linearity of Continuous Predictors in the Logit

A qualitative evaluation was conducted to comprehend the relationship between continuous predictors and the logit of the outcome. Typically, the Box-Tidwell test is the formal method for assessing linearity; it was not conducted here. Instead, an assumption was made that the predictors PCTBACHMOR (percentage of bachelor's degree or higher) and MEDHHINC (median household income) have approximately linear relationships in the logit.

The logistic regression results (e.g., the non-significance of PCTBACHMOR) suggest that these predictors have minimal practical impact.

Independence of Observations

In a logistic regression, the assumption of independence is important. However, this study had limitations in that there was no temporal sequence data to test time-based dependencies and lacked geographic coordinates to conduct spatial autocorrelation tests, such as Moran’s I. Despite this, the dataset structure assumes independence simply from an observation level. Crashes in clusters may violate this assumption, but the significance of these effects is not known without further testing.

Dataset Sample Size

The dataset contained 43,364 observations with 2,485 alcohol-involved crashes, which yields great statistical power. The event per variable (EPV) ratio is roughly 276, exceeding the 50 EPV ratio required for stable outcomes. Each binary predictor has a sufficient category-specific sample size, yielding reliable estimates.

Logit Regression Results

Figure 6 below presents the results of the logit regression from R.

Table 4

Logistic Regression Results & Odds Ratios

Predictor	β Coefficient	Std. Error	z-value	p-value	Odds Ratio	95% CI Lower	95% CI Upper
(Intercept)	-2.733	0.046	-59.56	<0.001	0.065	0.059	0.071
FATAL_OR_M	0.814	0.084	9.71	<0.001	2.257	1.910	2.653
OVERTURNED	0.929	0.109	8.51	<0.001	2.532	2.035	3.122
CELL_PHONE	0.030	0.198	0.15	0.881	1.030	0.684	1.488
SPEEDING	1.539	0.081	19.11	<0.001	4.660	3.974	5.450
AGGRESSIVE	-0.597	0.048	-12.49	<0.001	0.551	0.501	0.604
DRIVER1617	-1.280	0.293	-4.37	<0.001	0.278	0.148	0.471
DRIVER65PLUS	-0.775	0.096	-8.08	<0.001	0.461	0.380	0.553

The strongest positive predictors are the following:

- Speeding

  - Speeding had the highest positive association with alcohol-related crashes, with an odds ratio (OR) of 4.660 and $p < 0.001$. In other words, a crash involving speeding is 4.6 times more likely to involve alcohol than a crash where speeding was not a factor.
- Overturned Vehicles
  - This predictor has a strong positive association ($OR = 2.352$, $p < 0.001$), suggesting crashes involving overturned vehicles are 2.5 times more likely to involve alcohol.
- Fatal or Major Injuries
  - With an OR of 2.257 ($p < 0.001$), crashes with severe injuries are slightly more than twice as likely to involve alcohol compared to less severe crashes.

The significant negative predictors are the following:

- Age-Based Predictors
  - Both drivers aged 16-17 and 65+ showed negative associations with crashes. Crashes involving drivers 16-17 are 72% less likely ($OR = 0.278$, $p < 0.001$) to involve alcohol, whereas drivers 65+ are 54% less likely ($OR = 0.461$, $p < 0.001$) to be in crashes with alcohol compared to other drivers.
- Aggressive Driving
  - With an $OR = 0.551$ ($p < 0.001$), crashes involving aggressive driving was 45% less likely to involve alcohol compared to crashes where aggressive driving was not reported.

The nonsignificant predictors are the following:

- Cell Phone Use
    - With $p = 0.881$, cell phone usage did not show a statistically significant association with alcohol-related crashes. This means cell phone usage during crashes does not affect the likelihood of alcohol involvement.
- Percentage of Individuals w/ a Bachelor's Degree or Higher
    - With $p = 0.775$, there is no significant association with alcohol involvement, meaning educational attainment is irrelevant.

A negligible socioeconomic factor was median household income (MEDHHINC), which was statistically significant. However, the OR was close to 1.000. The impact on the likelihood of alcohol-related crashes was minimal in real terms.

Table 5
Model Fit Statistics

Model Statistic	**Outcome**
Null Deviance	19,036 on 43,363 degrees of freedom
Residual Deviance	18,340 on 43,354 degrees of freedom
AIC	18,360
Number of Fisher Scoring Iterations	6

The model reveals that most behavioral factors, especially speeding, are the strongest predictors of alcohol impairment in crashes. The severity of crashes, including whether a vehicle is overturned, also shows strong positive associations. Age predictors, both young and old, suggest that alcohol-related crashes are less common among the elderly and the young. Both education and income show minimal to no practical significance in predicting alcohol involvement.

A high residual deviance in the model suggests that although these predictors are significant, the current model does not capture other important factors that could help explain alcohol involvement in crashes.

Cut-Off Value Analysis

When conducting a cut-off value analysis, the probability threshold that yields the lowest misclassification rate is a cut-off value of 0.5, where the misclassification rate is 0.057.

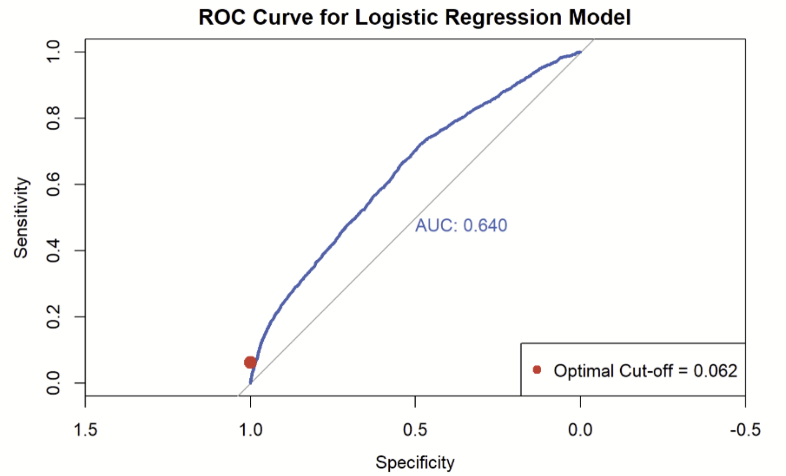
Table 6

Model Performance and Cut-Off Values

Cut-off Value	***Sensitivity***	***Specificity***	***Misclassification Rate***
0.02	0.983501006	0.05807383	0.88889401
0.03	0.980684105	0.06392035	0.88354395
0.05	0.734808853	0.46909171	0.51568121
0.07	0.221327968	0.91381883	0.12586477
0.08	0.184708249	0.93862374	0.10457984
0.09	0.168209256	0.94596247	0.09860714
0.1	0.164185111	0.94821302	0.09671617
0.15	0.104225352	0.97221067	0.07752975
0.2	0.022937626	0.99537660	0.06034960
0.5	0.001609658	0.99990215	0.05730560

Figure 6

ROC Curve for Logistic Regression Model



The optimal cut-off rate is selected by minimizing the distance from the upper left corner of the ROC curve, which yielded 0.640. The cut-off balances sensitivity and specificity by identifying the point closest to the ideal top-left corner of the ROC Curve, where sensitivity and specificity are maximized.

This value differs significantly from the typical threshold of 0.5, which assumes equal costs for false positives and negatives. Minimizing the distance from the upper left corner considers errors, optimizing the prediction performance. This approach is practical when the class distribution is imbalanced, since only 5.37% of crashes were alcohol-induced (Y=1), the dataset is heavily skewed toward non-alcohol-induced crashes (Y=0). Therefore, a lower cut-off value of 0.0637 was selected to improve sensitivity and detect additional alcohol-induced crashes while balancing specificity.

The model's relatively high residual deviance suggests that while these predictors are significant, there may be other important factors not captured in the current model that could help explain alcohol involvement in crashes.

The ROC curve in blue shows that it performs better than random allocation of accidents, regardless of whether they involve an alcohol-impaired driver, indicated by the dotted line in red. The area under the curve is 0.64. While this is better than random at 0.5, it is still a poor model. It lacks strong discriminatory power in distinguishing between alcohol-induced car crashes and car crashes that are not.

Another logistic regression model was created using only binary predictors, excluding PCTBACHMOR and MEDHHINC.

Table 7

Regression Model with Only Binary Predictors

Predictor	β Coefficient	Std. Error	z-value	p-value	Odds Ratio	95% CI Lower	95% CI Upper
(Intercept)	-2.652	0.028	-96.324	<0.001	0.071	0.067	0.074
FATAL_OR_M	0.809	0.084	9.662	<0.001	2.246	1.901	2.640
OVERTURNED	0.940	0.109	8.619	<0.001	2.559	2.057	3.156
CELL_PHONE	0.031	0.198	0.157	0.875	1.032	0.685	1.491
SPEEDING	1.540	0.081	19.128	<0.001	4.666	3.980	5.457
AGGRESSIVE	-0.594	0.048	-12.433	<0.001	0.552	0.503	0.606
DRIVER1617	-1.272	0.293	-4.338	<0.001	0.280	0.149	0.475
DRIVER65PLUS	-0.766	0.096	-8.004	<0.001	0.465	0.383	0.558

The magnitude of the coefficients in this model for FATAL_OR_M, DRIVER1617, and DRIVER65PLUS is smaller than that of the original model. FATAL_OR_M decreased from 0.814 in the first model to 0.809 in this model. This indicates that the socioeconomic variables could explain a tiny fraction of the crash characteristics in the initial model, since the change is miniscule. Including census-based socioeconomic factors reduced the direct influence of these crash-related predictors on alcohol involvement.

The new model (not including the census-based variables) had larger magnitudes for the coefficients of OVERTURNED, CELL_PHONE, and SPEEDING. Excluding socioeconomic variables in the new model caused these behavioral variables to take on more explanatory power. However, the difference is relatively small.

The Cell_PHONE variable is consistently insignificant in both models, where all other variables maintained their statistical significance.

Table 8
Akaike Information Criterion for Each Model

MODEL	AIC
ORIGINAL MODEL	18359.63
MODEL WITHOUT CENSUS VARIABLES	18360.47

The Akaike Information Criterion (AIC) of the first regression model with more parameters is 18359.63, whereas the AIC of the second regression model, excluding PCTABCHMOR and MEDHHINC, is 18360.47. The first regression, including the census variables, has a slightly lower AIC, which means it performs slightly better.

Discussion

` To determine whether a crash was alcohol-related, this report employed a logistic regression model incorporating various predictor variables. These included both driver demographics and census tract characteristics. The strongest predictors of alcohol involvement are speeding, vehicle overturning, and fatalities in the crash. In contrast, cell-phone usage showed no significant association, nor did any census tract variables (percentage of residents with a bachelor's degree or higher and median household income).

These findings align with expectations, but also present contradictions to others. Since impaired driving is paired with poor judgment, it is unsurprising that there is a notable link between alcohol-related crashes and speeding. However, contrary to original assumptions, aggressive driving behaviors were associated with a lower likelihood of alcohol involvement. This unanticipated result may be because aggressive driving is more closely related to road rage rather than inebriation. Additionally, a crash can occur at any point during the trip; the socioeconomic characteristics of the crash location (educational attainment and income levels) had no discernible impact.

Considerations for Rare Events in Logistic Regression

A key concern when modeling rare events in logistic regression is small-sample bias, which happens when the outcome of interest is infrequent. Only 5.73% (2,485 out of 43,364) of the crashes in the entire dataset are alcohol related. Even with this low rate, the absolute number of cases (2,485) can sufficiently minimize substantial bias in standard logistic regression. Thus, conventional methods remain appropriate for this analysis.

That said, penalized likelihood methods such as Firth's correction, as suggested by Paul Allison, are often recommended for rare events, mitigating bias and producing more stable estimates. Although adjustments may not drastically alter results (due to a high event count), they offer a more robust approach to ensure consistent and reliable parameter estimates.

Limitations

A notable limitation of this analysis is the restricted set of predictor variables. Adding additional factors, such as prior DUI/DWI offenses or time of day, could improve the model's predictive accuracy to provide deeper insights into impaired driving patterns. Future research should consider these variables to enhance model performance.